

AYE
TECH HUB

• CRITICAL THINKING SERIES • NO. 02

Cognitive Biases

The Errors Your Brain Makes Daily

Your Brain is Brilliant — and Brilliantly Wrong.

No. 02 of 30 | Critical Thinking Series

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- 20 cognitive biases explained with real-world examples
- Why your brain is wired to be wrong — and why that is OK
- The Dunning-Kruger curve — competence vs confidence
- How social media and AI exploit your cognitive weaknesses
- Practical strategies to de-bias your thinking today

Your Brain is Wired to Be Wrong

A **cognitive bias** is a systematic pattern of deviation from rationality in judgement. Your brain is not broken — it evolved these shortcuts (*heuristics*) over millions of years to make fast decisions with limited information. In the savannah, this worked brilliantly. In the information age, many of these same shortcuts actively mislead us.

There are over 180 documented cognitive biases. This issue covers the 20 most consequential — especially those **amplified by social media, AI, and digital information environments**. Knowing your biases does not eliminate them. But it gives you the pause, the moment of awareness, to choose differently.

"The first principle is that you must not fool yourself — and you are the easiest person to fool."

— Richard Feynman, physicist

The Four Categories of Cognitive Bias



Figure 1 — The Four Categories of Cognitive Bias

The Dunning-Kruger Effect — The Most Dangerous Bias

Described by psychologists David Dunning and Justin Kruger in 1999, this effect shows that people with **low competence in a subject dramatically overestimate their ability**, while genuine experts tend to underestimate theirs. The curve below is arguably the most important graph in all of cognitive psychology.

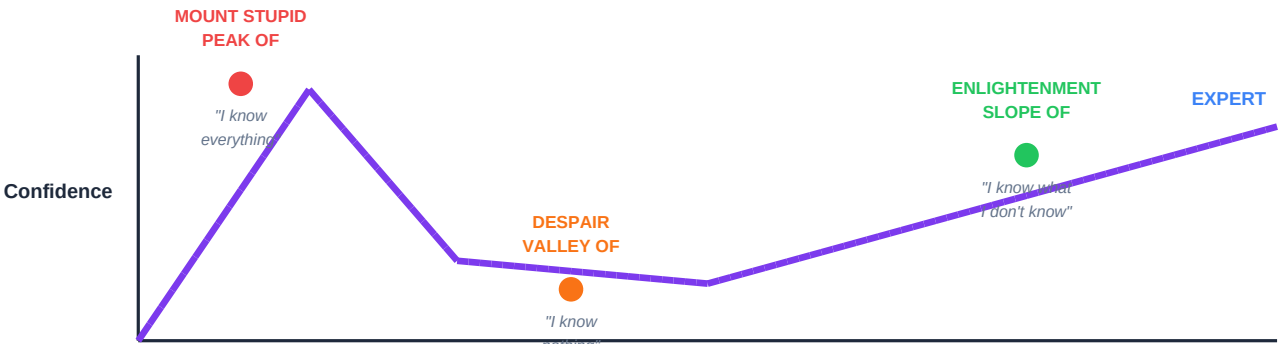


Figure 2 — The Dunning-Kruger Effect: Confidence vs Competence

AI age application: After a few ChatGPT conversations about medicine, law, engineering, or finance, many people reach the "Peak of Mount Stupid" — confident they now understand the field. The cure is simple: keep learning until you reach the Valley of Despair, then climb the Slope of Enlightenment.

The 10 Most Dangerous Biases — Deep Dive

The following ten biases are responsible for the majority of poor decisions, failed relationships, bad investments, political polarisation, and misinformation spread. Study each one — and ask yourself honestly: where did this bias affect me last week?

Bias	What It Does	Real-World Trap	Counter-Move
1. Confirmation Bias	You seek, notice and remember only information that confirms what you already believe	You only follow news sources, people and AI outputs that agree with your worldview	Actively seek the best argument AGAINST your position before deciding
2. Availability Heuristic	You judge how likely something is based on how easily an example comes to mind	You think plane crashes are common (vivid news coverage) but car crashes are rare	Ask: am I estimating probability from statistics or from memorable examples?
3. Anchoring Bias	You rely too heavily on the first piece of information you encounter	A salary negotiation starts at a low anchor, pulling the final number down	Generate your own estimate BEFORE seeing any external number
4. Sunk Cost Fallacy	You continue a failing course of action because of past investment (time, money, emotion)	Staying in a bad job, relationship or project because "I have already invested so much"	Ask: if I had zero past investment, would I start this today?
5. Hindsight Bias	After an event, you believe you "knew it all along" — even if you did not	Market crashes, election results and failures all seem "obvious in retrospect"	Keep a decision journal before events to track your actual predictions
6. Framing Effect	The same information presented differently produces different decisions	"90% survival rate" vs "10% death rate" — same fact, completely different reactions	Reframe every important decision in both positive and negative terms
7. Attribution Error	You attribute others' behaviour to their character, not their situation	Someone is rude to you — you think "they are a bad person", not "they had a terrible day"	Ask: what external pressures might be driving this behaviour?
8. Bandwagon Effect	You adopt beliefs and behaviours because many others have	Cryptocurrency bubbles, viral trends, viral misinformation, group consensus	Ask: would I believe this if no one else did? What is the actual evidence?
9. Optimism Bias	You overestimate the probability of positive outcomes for yourself	"I will not get cancer." "My startup will succeed." "This project will finish on time."	Use reference class forecasting: how often do similar projects actually succeed?
10. Automation Bias	You over-trust outputs from automated or AI systems without critical evaluation	Accepting GPS directions into a lake. Submitting AI content without fact-checking.	Treat AI output as a first draft, not a final answer. Always verify key claims.

The confirmation bias loop: Of all the biases above, confirmation bias is the most dangerous in the AI age — because algorithms are specifically designed to feed it.

The Confirmation Bias Echo Chamber

Confirmation bias does not just affect individuals — it is the engine that powers **social media algorithms, political polarisation, conspiracy theories, and AI filter bubbles**. When you only see content that confirms your existing beliefs, your beliefs become more extreme, more rigid, and more detached from reality — without you ever noticing.

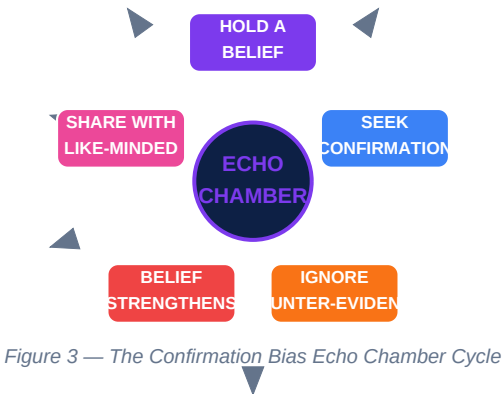


Figure 3 — The Confirmation Bias Echo Chamber Cycle

The Next 10 Biases — Know Their Names

Bias	The Core Error	AI Age Example
11. Status Quo Bias	Preferring the current state of affairs; change feels like loss	Refusing to update your skills because "the old way works fine"
12. In-Group Bias	Favouring people from your own group (nationality, religion, team)	Only trusting AI tools from companies you already like
13. Recency Bias	Giving excessive weight to recent events over longer-term trends	Assuming the latest AI model is always the best one for every task
14. Survivorship Bias	Focusing on successes while ignoring the failures you never hear about	"This entrepreneur dropped out of school and got rich" — ignoring thousands who did not
15. Planning Fallacy	Underestimating time, costs and risks of future projects	Every AI project will take longer and cost more than the initial estimate
16. False Consensus Effect	Overestimating how much other people share your beliefs and values	Assuming everyone finds the AI tools you love to be as intuitive as you do
17. Negativity Bias	Negative events have greater emotional impact than equally positive ones	One critical comment about your work weighs more than ten pieces of praise
18. Spotlight Effect	Overestimating how much others notice and think about you	Afraid to share your ideas publicly because "everyone will judge me"
19. Mere Exposure Effect	You like things more simply because you have been exposed to them more	Trusting a specific AI brand because you see it advertised constantly
20. Just-World Hypothesis	Believing the world is fundamentally fair — good things happen to good people	Blaming victims of injustice for their own misfortune

How to Start De-Biasing Your Thinking

Strategy	How It Works	When to Use
Pre-mortem analysis	Before a decision, imagine it has failed. Ask: why did it fail?	Before major decisions, projects, investments
Seek disconfirmation	Actively search for the strongest argument against your position	Anytime you are forming a strong opinion on something important

Consider the base rate	What is the historical success/failure rate for situations like this?	Estimating probabilities, planning projects, career decisions
Name the bias out loud	When you catch a bias in action, naming it reduces its power	In real-time decision making and group discussions
Slow down the System 1	Gut reactions are fast (System 1). Deliberate thinking is slow (System 2). Pause.	Any time you feel a strong emotional pull toward a conclusion
Get a thinking partner	Find someone who will honestly challenge your conclusions, not validate them	For important decisions — career, relationships, business, beliefs

What comes next — CT-003: How to Spot Misinformation and Fake News — a practical field guide for navigating the information war, with checklists, source evaluation frameworks, and the five types of misinformation you will encounter every single day.